



ENGINEERING PROFESSION IN NIGERIA: PATHWAYS, OPPORTUNITIES, RESPONSIBILITIES, IMPACT & THE FUTURE

A paper delivered by
Engr. Ayodeji Ajayi MNSE, MNIEEE
at the
Engineers-in-Society Symposium
Held on: Tuesday, 24th February 2026
Venue: Hilltop Auditorium, Obanla, Federal
University of Technology, Akure

Presentation Outline

What is Engineering/Who is an Engineer?

The Role of Engineers in Society

Engineering Ethics and Responsibilities

Engineers Impact on National Development

Challenges facing Engineers in Nigeria

Emerging Trends in Engineering

Skills Required to Succeed as an Engineer

What Students Must do Now

Professional Engineering Pathway

Conclusion

+
•
◦

What is Engineering/ Who is an Engineer?

An engineer applies scientific principles to design, build, and maintain systems that improve life

- Problem-solving profession
- Driver of innovation
- Societal transformation
- A designer of systems, structures, and solutions
- A custodian of public safety
- A driver of national development

The Role of Engineers in Society

- **Infrastructure Development**

- Roads, bridges, water systems, housing, power generation and distribution, transportation systems

- **Industrial & Economic Growth**

- Manufacturing, Mining, Oil & gas, ICT and automation

- **Innovation & Technology**

- Renewable energy, Artificial intelligence, Smart cities, Sustainable mining, IoT & IIoT, Telecommunications,

- **Public Safety & Standards**

- Compliance with engineering codes, Ethical decision-making, Risk assessment, Environmental Protection



Engineering Ethics and Responsibility



The NSE Code of Conduct emphasizes:

- ✓ Integrity
- ✓ Competence
- ✓ Professional responsibility
- ✓ Respect for life and environment
- ✓ Accountability
- ✓ Safety first
- ✓ Public interest
- ✓ Environmental responsibility

The Engineer's Impact on National Development



Engineers shape the physical and economic landscape of a nation.



No country develops without strong engineering capacity.



Nigeria's future depends on young engineers who can innovate, adapt, and lead.

Challenges Facing Engineers in Nigeria

Inadequate infrastructure

Funding limitations

Technological gaps

Policy inconsistencies

Brain drain

Emerging Trends in Engineering

Automation & robotics

Renewable energy systems

Smart mining technologies

Artificial intelligence and Machine Learning

6G Technology (Holographic and Immersive Communication)

Edge Computing

Quantum Computing (Qubits)

Blockchain Technology

Sustainable development practices

Professional Engineering Pathway

- Undergraduate Training (Student Membership)
- NYSC (Graduate Membership)
- Industrial Experience (Graduate Membership)
- Professional Registration (Nigeria Society of Engineers Corporate Membership)
- Continuing Professional Development (CPD) (Training, Seminars and Conferences)
- COREN Registration (Council for the Regulation of Engineering in Nigeria – Your license to practice)

Skills Required to Succeed as an Engineer

- Technical Skills
- Strong mathematics & analysis
- Software proficiency
- Design tools
- Soft Skills
- Communication
- Leadership
- Teamwork
- Ethics
- Critical thinking

What must I do as a student?

- Build strong fundamentals
- Engage in practical projects
- Join professional bodies (NSE, NICE, NIEEE, NSChE, NIMechE,)
- Seek mentorship
- Develop communication & leadership skills
- Embrace continuous learning
- Take academics seriously
- Learn software early
- Seek internships
- Work on practical projects

Conclusion

- ❑ Engineering is more than a degree; it is a calling to serve humanity.
- ❑ Your ideas, your creativity, and your commitment will shape the Nigeria of tomorrow
- ❑ Specialization matters
- ❑ Skills development is continuous
- ❑ Integrity defines success
- ❑ Nigeria needs competent engineers
- ❑ There is an urgent need to bridge the gap between the academic and the practicing Engineers

Parting Words

1

Don't just pass exams, understand the principles.

2

Join student chapters of NSE/NIMechE/NIEE E. NICE and so on.

3

Be curious. The classroom is just the beginning.

4

Be the solution which the country really desires.

The background features several abstract geometric shapes in dark gray: a large semi-circle on the right, a solid circle to its left, a dashed curved line below the solid circle, a square outline on the far left, a vertical dashed line above the square, a checkmark-like shape above the large semi-circle, and a semi-circle at the top right.

**Thank you for
listening**